



A Simultaneous Planning and Control Method Integrating APF and MPC to Solve Autonomous Navigation for USVs in Unknown Environments

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Abstract

This paper is devoted to solving autonomous navigation for unmanned surface vessels (USVs) in unknown environments. To overcome the deficiency of the “first planning then tracking” motion control framework, a novel simultaneous planning and control (SPC) method is developed. The developed method combines an improved artificial potential field (IAPF) and model predictive control (MPC) techniques. Improvements in the IAPF are made to deal with constraints on angular velocity. In each step of the SPC method, the IAPF is used for robust and efficient tracking in a short future. And the MPC is implemented to generate actual control commands for high-precision tracking. The IAPF and the MPC work in an alternative way to drive the USV to the prescribed target while avoiding the obstacles detected around. Simulations with static and dynamic obstacles demonstrate the effectiveness of the proposed method. The method works well when maneuvering in complex environments even crossing narrow tunnels.

Keywords Unmanned surface vehicles · Motion control · Artificial potential field · Model predictive control · Simultaneous planning and control

1 Introduction

1.1 Background

Unmanned surface vessel (USV) is a kind of water-borne vessel without pilot, owing the advantages of small volume,

light weight, low cost and strong adaptability. Initial applications of USVs trace back to World War II when remote-control vessels are used to attack enemy vessels. Nowadays, USVs have been widely used in various scenarios, including both civil engineering and military operations, especially for tasks with high risk, long working time and hostile environment. However, there are still some crucial technical problems that need to be further studied to realize robust autonomous navigation for USVs. Related issues, as illustrated in Fig. 1, can be categorized into environment perception, path planning, motion control and state/parameter estimation.

Environment perception refers to recognition of operational environment and obstacles surrounding, and it's mainly supported by pattern recognition and information fusion. Based on the obtained environment information, path planning module generates an optimal path for the USV. Motion control module will provide control commands to make the USV move along with the planned route safely and reliably in the presence of external disturbances, it covers kinematic/dynamic modeling and automatic control theories. State estimation (sometimes including parameter as well) provide USV's accurate own state (such as position, posture,

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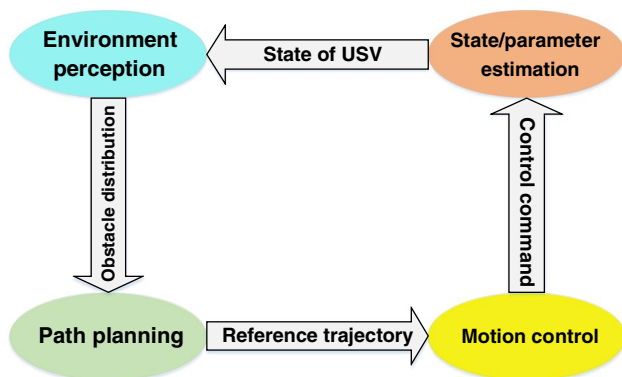


Fig. 1 Four crucial issues for USV autonomous navigation

velocity, etc.). As in this paper, we focus on path planning and motion control issues involved.

1.2 Literature Review

The path planning of USV is to find a path from the current state to the desired target. Generally speaking, two important factors need to be dealt with carefully. On the one hand, a reasonable objective should be designed to evaluate the effectiveness or reliability of the path. And typical objectives include shortest path, time-optimal, minimum energy, etc. On the other hand, collision avoidance should be achieved so that the USV can safely accomplish the mission.

Similar to that of other vehicles, path planning algorithms of USVs are grouped into global and local path planning. Global path planning methods mainly include curve fitting [1, 2], graph-based method [3–5], intelligent algorithm [6–8], etc. And local methods mainly include artificial potential field (APF) method [9–11], dynamic window method [12], optimal control method [13, 14], etc. According to whether time information is involved in the obtained path, path planning techniques can be also classified into geometry methods and trajectory planning methods. In most geometry methods, a UAV is seen as a mass point. However, actual USVs are essentially complicated underactuated systems with dynamical constraints. Hence, it's quite questionable whether the path generated by geometry methods can be successfully tracked in reality. Optimal control, as a trajectory planning method, can fully consider the dynamical maneuverability characteristics of the USV and various constraints effectively under a uniform framework. A nonlinear optimization problem is required to be solved, where involved nonlinear motion equations and collision-avoidance constraints make the feasible region of the optimization problem extremely non-convex. And one must carefully set the initial guess of the optimization problem to facilitate a robust numerical solution.

When a reference path is obtained, the following issue, as the key step to realize autonomous navigation, is to control the USV to track reference trajectory. Due to the uncertainty of external environment, the actual trajectory of the USV may deviate from the reference trajectory, which is particularly more apparent for USV than vehicles on land. Wind and wave will affect the motion control performance of USV, and the maneuverability varies according to speed, sailing depth and carrying capacity. An experienced pilot can adjust the operation strategy to deal with these situations. However, for USVs, reliable onboard trajectory tracking controllers must be equipped.

Related control techniques include backstepping method [15, 16], fault-tolerant tracking control [17, 18], sliding mode control [19, 20], adaptive control [21–23], intelligent method [24] and model predictive control (MPC) [25–27], etc. Although these USV trajectory tracking control methods have achieved some achievements, it remains a wide investigation to realize real-time and high-precision trajectory tracking of USV under control-state constraint with external disturbance.

In the traditional autonomous navigation framework for USVs, a reference path is planned offline first and used for tracking, leading to a “first planning then tracking (FPTT)” framework. For example, Bergman et al. [28] develop a bi-layer motion controller, where a lattice-based motion planner is used in the first layer to generate a feasible trajectory. This trajectory is then used to enable a warm-start and as a terminal manifold for the MPC in the second layer. And they further integrate the COLREGs rules into the motion planner in the first layer to improve the maneuver safety [29]. To improve the planning performance, Bitar et al. [30] incorporate hybrid A* and optimal control algorithm. However, both the lattice-based method and hybrid A* method exhibits low search efficiency when USV crosses narrow tunnels. It should be noted that such an FPTT framework requires the whole information on the global environment before executing the task. However, when sailing in an unknown environment, it is hard to obtain the global terrain information in advance. Additionally, the path planning module and the trajectory tracking module are totally separated. If the USV deviates from the prescribed path severely under external disturbances, this planned path fails. Then a path re-planning is required, and it would be extremely dangerous for USVs during this re-planning period.

1.3 Motivations and Contributions

From the above discussions, it is seen that the FPTT framework has its inherent drawbacks. To achieve autonomous navigation in unknown environments, the USV must continuously detect the surrounding environment to construct a temporary map for path planning, followed by an immediate

trajectory tracking. We come up with a simultaneous planning and control (SPC) framework, we the local path planning module and the trajectory tracking module works in an alternative way. To achieve efficient and robust motion planning, the APF method is selected. The APF method was first proposed by Khatib [31] to solve the path planning problem of manipulators. As the advantages of high computational efficiency and sound capability in obstacle avoidance, the APF has been widely used and many improved versions have been developed. However, the time information is still rarely provided in these modified algorithms, making it hard to conduct the following trajectory tracking. Moreover, most of them don't consider the maneuverability of the USV. To improve the quality of APF-generated paths, we make some novel attempts and thus develop an improved APF (IAPF) method. Due to the existence of external disturbances, it is not necessary to generate a complete path from the current position to the target. Actually, when a reference trajectory in the short future is generated, a trajectory tracking controller can be immediately applied. The tracking controller based on the dynamics of the USV can compute actual inputs to drive it to sail along with the reference trajectory without severe deviation. To deal with necessary constraints on state and control variables, MPC, as a closed-form extension of optimal control techniques, is adopted in this paper. Since collision avoidance is achieved by the IAPF, collision-free constraints can be eliminated in the design of MPC. It results in two merits. On the one hand, it overcomes the deficiency when optimal control method encounters obstacles in complex shapes; on the other hand, it guarantees the online computational efficiency. The SPC framework integrates the advantages of both IAPF and MPC. Detailed, IAPF can generate high-quality reference trajectory in milliseconds, while MPC considers various constraints and deal with external disturbances.

The main contributions of the paper are summarized as follows:

- (1) An IAPF with two major modifications is developed for implementing fast path planning. On the one hand, information on time is added into APF to facilitate the following trajectory tracking. On the other hand, the constraint on angular velocity is considered to remove the oscillating behavior in APF when detouring obstacles.
- (2) The MPC tracking module fully considers various state and control constraints on USVs. The collision avoidance is not imposed when designing the MPC tracking controller since a collision-free reference trajectory is given in each step, significantly improving the online computational efficiency.
- (3) In the developed SPC framework, by using only local terrain information within the perception range, the

IAPF and the MPC cooperate alternatively to realize autonomous navigation in unknown environments. No complete reference trajectory is required before conducting the trajectory tracking, guaranteeing the real-time control performance. The method works well even when the USV maneuver across an obstacle-dense environment.

1.4 Organization

The remainder of the paper is organized as follows. Section II gives the mathematical model of the underactuated USV. An IAPF method with constraints on angular velocity is introduced and tested in Section III. To make the generated trajectory dynamically feasible with dynamics and various practical constraints of USV, an SPC method is developed in Section IV. Section V gives two examples to validate the developed method. Finally, Section VI concludes the paper.

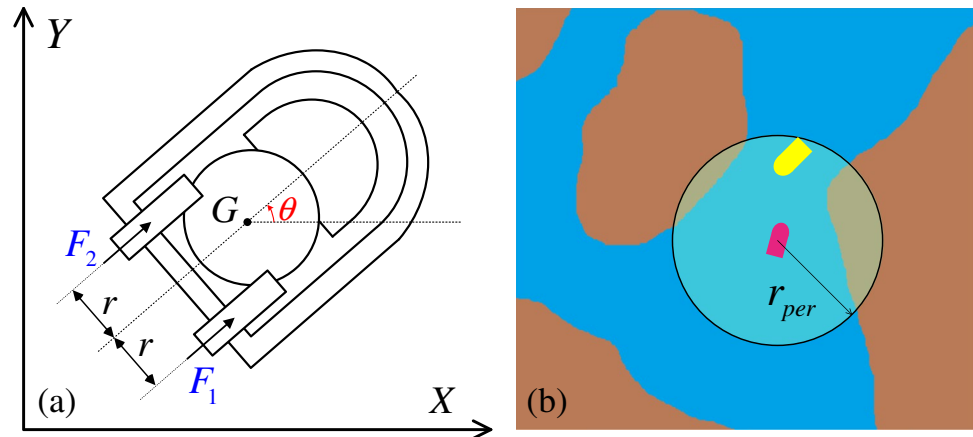
2 Mathematical Model of the Underactuated USV

2.1 Motion Equation

Consider the USV illustrated in Fig. 2(a), where Point $G(x, y)$ is the mass center and θ represents the orientation. The velocity and the angular velocity of the USV are denoted as v and ω , respectively. The USV is equipped with two parallel propellers, and the distance between a propeller to the symmetry axis of the USV is r . The mass and the moment of inertia of the USV are denoted as m_c and I_c , respectively. The rate of change of propulsions is actually restrained due to mechanical constraints. To consider such a constraint, we view the two propulsions as extra state variables and view rates of change of propulsions as the control variables. Hence, the motion of the USV can be described by the following equations

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}, t) = \begin{bmatrix} v \cos \theta \\ v \sin \theta \\ \omega \\ (F_1 + F_2)/m_c \\ (F_1 - F_2)r/I_c \\ \dot{F}_1 \\ \dot{F}_2 \end{bmatrix} \quad (1)$$

where $\mathbf{x} = (x, y, \theta, v, \omega, F_1, F_2)^T$ is the state space and $\mathbf{u} = (u_1, u_2)^T = (\dot{F}_1, \dot{F}_2)^T$ is the control variable list.

Fig. 2 Configuration and perception of the USV

2.2 Perception Ability

To enable the USV to navigate itself in an unknown environment, the USV must receive terrain information from other sources or carry certain equipment (radar, vision perception, etc.) to detect the obstacles around, including land and other vehicles. In this paper, as illustrated in Fig. 2(b), it is assumed that the USV can perceive the local environment within the radius of r_{per} .

2.3 Constraints

Both propulsions and their rates of change are restrained due to practical mechanical constraints, leading to the following inequality constraints

$$F_{\min} \leq F_i \leq F_{\max}, (i = 1, 2) \quad (2)$$

$$|u_i| \leq u_{\max}, (i = 1, 2) \quad (3)$$

where $[F_{\min}, F_{\max}]$ is the allowable range of the propulsions and $u_{\max}$ is the max allowable rate of change of the propulsions.

In addition, one should restrain the velocity and the angular velocity of the USV within a safety range in case of emergency brakes. Such constraints are defined as follows

$$0 \leq v \leq v_{\max} \quad (4)$$

$$|\omega| \leq \omega_{\max} \quad (5)$$

where $v_{\max}$ is the maximum allowable velocity and $\omega_{\max}$ is the maximum allowable angular velocity.

3 An Improved APF Method with Constraints on Angular Velocity

3.1 The Default Artificial Potential Field Method

In APF methods, the vehicle is seen as a mass point, and one designs an attractive and a repulsive potential field to navigate the vehicle. The attractive potential field attracts the vehicle to move towards the target while the repulsive one repulses the vehicle away from obstacles. As a classical path planning method, the APF method has been developed over decades with different improvements on the design of potential fields. In this paper, we take the most state-of-art APF method as the basic algorithm, namely the DAPF method.

To implement the APF method, the environment is decomposed into cells where each cell is assigned with a value of either 0 or 1. For a cell with the value of 0, it is occupied by an obstacle and thus results in a repulsive force to the vehicle. The mesh density of the cells should be carefully set to meet the balance between computational efficiency and representation accuracy for the actual environment. For the simplicity of derivations, we define the notation $\rho(A, B)$ to denote the Euclidean distance between Point A and Point B; and the unit vector that points to Point A from Point B is denoted as $\vec{i}_{BA}$.

3.1.1 Attractive Potential Field

The coordinate of the target is set as $V(x_g, y_g)$. Let $\rho(G, V)$ represent the distance between the USV and the target. Thus, the attractive potential field U_{att} is given as

$$U_{att} = \begin{cases} \frac{1}{2} K_{att} \rho^2(G, V) & , \rho(G, V) \geq \rho_g \\ K_{att} \rho(G, V) \rho_g = \frac{1}{2} K_{att} \rho_g^2 & , \rho(G, V) < \rho_g \end{cases} \quad (6)$$

where K_{att} is the attractive coefficient; ρ_g is the parabolic range that divides the attractive potential into a piecewise function. The attractive force $\mathbf{F}_{att}$, which is the minus gradient of U_{att} , can be derived as

$$\mathbf{F}_{att} = -\nabla U_{att} = \begin{cases} K_{att} \rho(G, V) \vec{i}_{GV}, & \rho(G, V) \geq \rho_g \\ K_{att} \rho_g \vec{i}_{GV}, & \rho(G, V) < \rho_g \end{cases} \quad (7)$$

3.1.2 Repulsive Potential Field

For the j th occupied cell that locates at $W_j(x_{W_j}, y_{W_j})$, the repulsive potential field is designed as

$$U_{rep,j} = \begin{cases} \frac{1}{2} K_{rep} \left(\frac{1}{\rho(G, W_j)} - \frac{1}{\rho_o} \right)^2 \rho^n(G, V), & \rho(G, W_j) < \rho_o \\ 0, & \rho(G, W_j) \geq \rho_o \end{cases} \quad (8)$$

where K_{rep} is the repulsive coefficient; ρ_o is the effective range of the repulsive potential field; n is the repulsive decay of the target. When applying the APF to navigation of the USV, ρ_o is required to be no larger than r_{per} due to the limited perception range. The corresponding repulsive force can be determined as

$$\mathbf{F}_{rep,j} = -\nabla U_{rep,j} = \mathbf{F}_{rep1,j} + \mathbf{F}_{rep2,j} \quad (9)$$

where

$$\mathbf{F}_{rep1,j} = \begin{cases} K_{rep} \left(\frac{1}{\rho(G, W_j)} - \frac{1}{\rho_o} \right) \frac{\rho^n(G, V)}{\rho^2(G, W_j)} \vec{i}_{W_j G}, & \rho(G, W_j) < \rho_o \\ 0, & \rho(G, W_j) \geq \rho_o \end{cases} \quad (10)$$

$$\mathbf{F}_{rep2,j} = \begin{cases} \frac{n}{2} K_{rep} \left(\frac{1}{\rho(G, W_j)} - \frac{1}{\rho_o} \right)^2 \rho^{n-1}(G, V) \vec{i}_{GV}, & \rho(G, W_j) < \rho_o \\ 0, & \rho(G, W_j) \geq \rho_o \end{cases} \quad (11)$$

Thus, the resultant force $\mathbf{F}_{res}$ is written as $\mathbf{F}_{res} = \mathbf{F}_{att} + \sum_{s=1}^S \mathbf{F}_{rep,s}$, where S is the number of occupied cells within the perception range. In Fig. 3, we take a simple scenario where only one occupied cell exists to illustrate the design of potential fields.

Consider at the k th iteration, the vehicle locates at $(x^{[k]}, y^{[k]})$, and orientation of the resultant force $\mathbf{F}_{res}^{[k]}$ is $\hat{\theta}^{[k]}$. It is assumed that the vehicle moves along the orientation $\hat{\theta}^{[k]}$ within the k th iteration. Once the step size is fixed as l , then the initial position of the vehicle at the $(k+1)$ th iteration becomes

$$\begin{cases} x^{[k+1]} = x^{[k]} + l \cos \hat{\theta}^{[k]} \\ y^{[k+1]} = y^{[k]} + l \sin \hat{\theta}^{[k]} \end{cases} \quad (12)$$

Remark 1 Recent improvements have been made to deal with dynamic obstacles in APF methods. Actually, we can categorize dynamic obstacles into two groups. The first group is the cooperative obstacle whose future state information (including position, orientation, velocity, etc.) is available to the vehicle at the current moment. A typical example is the leader to the followers in the formation keeping control problem. For this kind of dynamic obstacles, relative velocity/orientation between the vehicle and obstacles can be utilized to design modified repulsive potential fields. The second group is the non-cooperative obstacles whose future state information is unavailable. One of the typical scenarios is the mobile target capturing problem. Generally, we treat this kind of obstacle by simply assuming the obstacle moves at the current velocity and orientation in this iteration or further integrating state prediction techniques.

3.2 An IAPF Method

It is noted that in the DAPF, the vehicle is assumed able to turn to the newly generated orientation immediately. However, the vehicle is essentially a rigid body subjected to dynamics rather than a mass point. Hence, the “immediate turning” behavior in the DAPF cannot be achieved in practice. A simple but efficient idea to deal with this deficiency is to introduce the constraints on angular velocity. In addition, information on time is not provided in the DAPF method. It results in the DAPF method being a path planning method rather than a trajectory planning method, making it impossible to implement

Fig. 3 Illustration of the resultant force in the APF

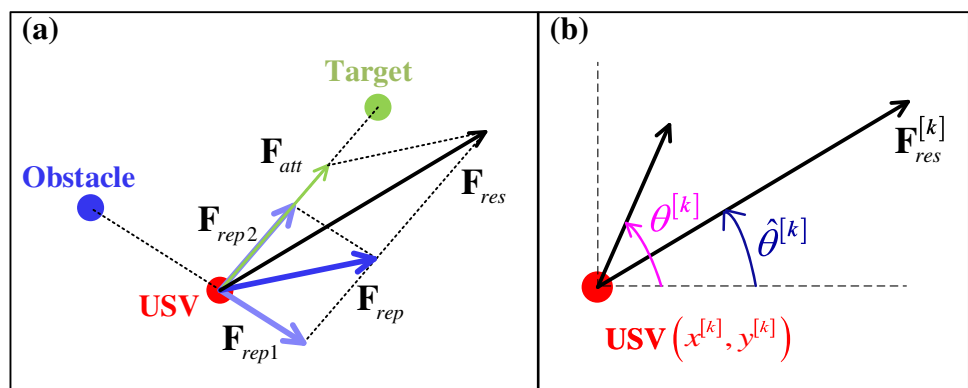
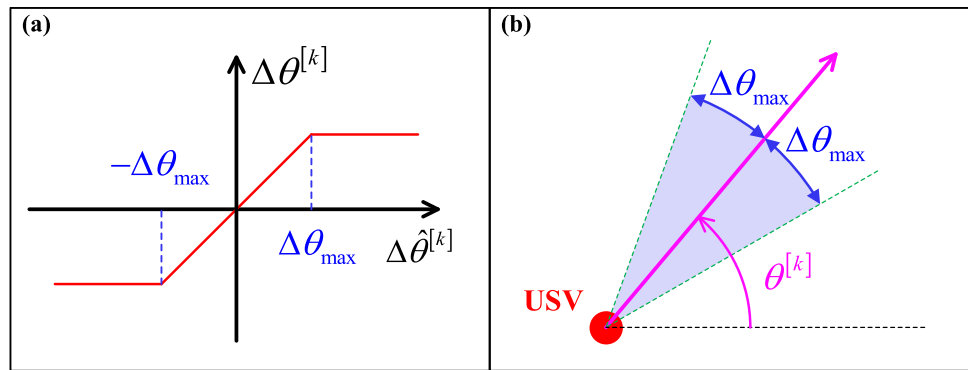


Fig. 4 The saturation function of variation of orientations in two adjacent iterations



trajectory tracking techniques. Actually, one can substitute the fixed step length in the DAPF method by the product of velocity of the vehicle and the fixed sampling interval. Based on the above ideas, we develop an IAPF method.

Assume the initial time is T_0 , and corresponding initial position and posture are $(x^{[0]}, y^{[0]})$ and $\theta^{[0]}$, respectively. In the IAPF, the k th sampling instant is defined as $T_k = T_0 + k \cdot \Delta T$ where ΔT is the fixed time step between two adjacent iterations. Given the vehicle moves at a constant velocity v_{const} , the fixed step size is equivalent to $\Delta T \cdot v_{\text{const}}$. By setting the maximum allowable angular velocity $\tilde{\omega}$, the maximum allowable change of orientation in two adjacent iterations is determined as $\Delta\theta_{\text{max}} = \Delta T \cdot \tilde{\omega}$. Define the following two notations

$$\Delta\theta^{[k]} = \theta^{[k+1]} - \theta^{[k]} \quad (13)$$

$$\Delta\hat{\theta}^{[k]} = \hat{\theta}^{[k]} - \theta^{[k]} \quad (14)$$

then, the orientation of the vehicle in the next iteration is determined according to the saturation function illustrated in Fig. 4. For simplicity, we define such a saturation function as

$$\begin{aligned} \Delta\theta^{[k]} &= \text{Saturation}(\Delta\hat{\theta}^{[k]}; \Delta\theta_{\text{max}}) \\ &= \min\{\max\{-\Delta\theta_{\text{max}}, \Delta\hat{\theta}^{[k]}\}, \Delta\theta_{\text{max}}\} \end{aligned} \quad (15)$$

Based on the above definitions, the initial position and posture information at the $(k+1)$ th iteration can be determined as Eq. (16). Together with the time information, the trajectory of the vehicle is obtained.

$$\begin{cases} x^{[k+1]} = x^{[k]} + \Delta T \cdot v_{\text{const}} \cdot \cos\theta^{[k+1]} \\ y^{[k+1]} = y^{[k]} + \Delta T \cdot v_{\text{const}} \cdot \sin\theta^{[k+1]} \\ \theta^{[k+1]} = \theta^{[k]} + \text{Saturation}(\Delta\hat{\theta}^{[k]}; \Delta\theta_{\text{max}}) \end{cases} \quad (16)$$

3.3 Validation of the IAPF Method

We consider an environment with a scale of $500 \text{ m} \times 400 \text{ m}$ and it is divided into cells with the size of $1 \text{ m} \times 1 \text{ m}$ to apply the APF methods. The vehicle departs from $(10 \text{ m}, 150 \text{ m})$ with an orientation of 0 deg. And the target is set as

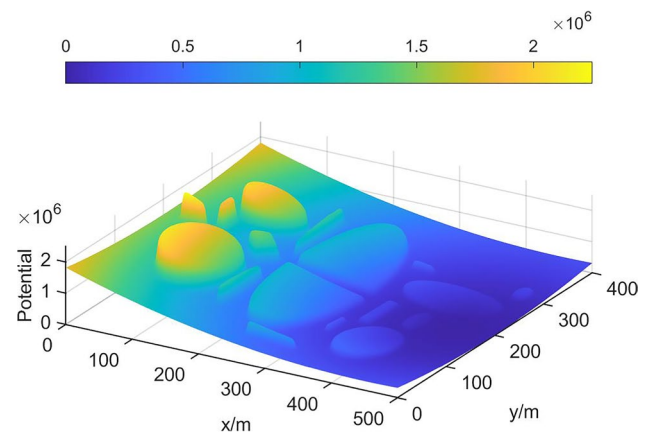


Fig. 5 The potential field

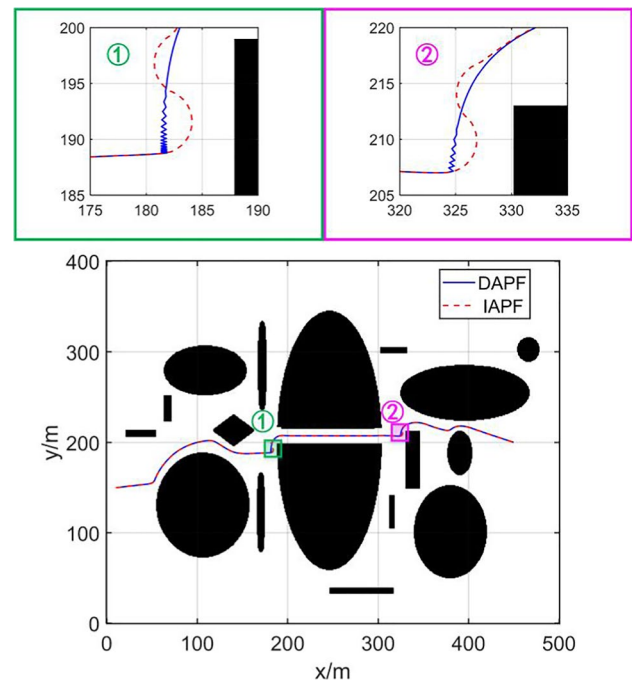


Fig. 6 Trajectories obtained by the DAPF and the IAPF

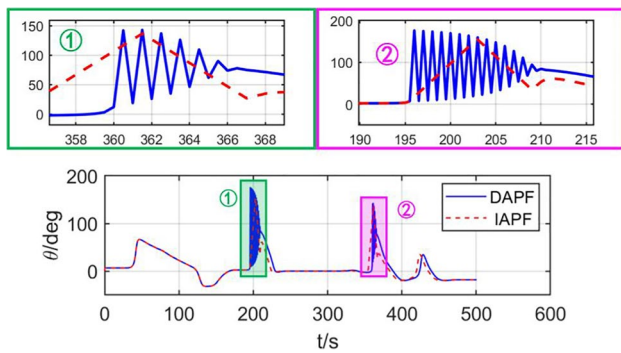


Fig. 7 Profiles of orientation

(400m, 200m). The algorithms parameters are set as $T_0 = 0s$, $\Delta T = 0.5s$, $v_{const} = 1m/s$, $\tilde{\omega} = 20 \text{ deg/s}$, $K_{att} = 15$, $K_{rep} = 5$, $\rho_g = 500m$, $\rho_o = 20m$ and $n = 2$. The potential field is plotted in Fig. 5, and the trajectories obtained by the DAPF and the IAPF are given in Fig. 6.

Though both the DAPF and the IAPF can successfully navigate the vehicle to the target by either rounding the obstacle or going through a narrow gap, we found the

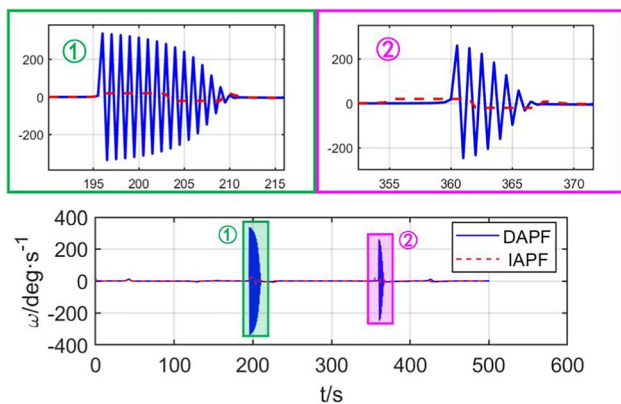


Fig. 8 Profiles of angular velocity

IAPF efficiently eliminates the oscillation when turning around obstacles as occurred in the DAPF. To demonstrate the advantages of the IAPF over DAPF, profiles of orientation and angular velocity are reported in Figs. 7 and 8, respectively.

Additionally, the properties of paths planned by the DAPF and the IAPF with different values of $\tilde{\omega}$ are summarized in Table 1. In Table 1, the variable $\Xi(\kappa)$ represents the number of iterations whose amplitude of angular velocity exceeds $\kappa \text{ deg/s}$, and the variable γ refers to the number of sign changes in angular velocity. Combined Table 1 and Figs. 7 and 8, it is seen that the paths generated by the IAPF and the DAPF show no great difference in length, and their computational efficiencies are almost equal. The IAPF can strictly restrain the angular velocity within the prescribed range, i.e., $[-\tilde{\omega}, \tilde{\omega}]$. However, as for the DAPF, frequent severe orientation oscillation occurs, which may arise significant potential risks.

4 A Simultaneous Planning and Control Method

4.1 Basic Idea

The IAPF can navigate the vehicle to the target successfully meanwhile satisfy the constraints on angular velocity. However, the IAPF cannot be directly applied in the autonomous navigation of USVs. On the one hand, the actual control commands (i.e., two propulsions or their rates of change) cannot be obtained. On the other hand, as observed from Eq. (16), it is actually designed based on quite simplified kinematics. Hence, it is questionable whether the generated trajectory is dynamically feasible since dynamics and various practical constraints are not taken into consideration. In this section, a simultaneous

Table 1 Comparison between paths generated by DAPF and IAPF

Property of the path	IAPF ($\tilde{\omega}=20 \text{ deg/s}$)	IAPF ($\tilde{\omega}=40 \text{ deg/s}$)	IAPF ($\tilde{\omega}=60 \text{ deg/s}$)	DAPF (no limit on $\tilde{\omega}$)
Number of steps	993	987	985	1001
Path length (m)	496.5	493.5	492.5	500.5
Total runtime (s)	4.59	4.49	4.86	4.58
Runtime per step (ms)	4.62	4.53	4.93	4.58
Maximum amplitude of angular velocity (deg/s)	20	40	60	338.31
$\Xi(10)$	63	37	31	33
$\Xi(20)$	0	28	19	23
$\Xi(40)$	0	0	16	21
$\Xi(60)$	0	0	0	17
γ	7	7	11	43

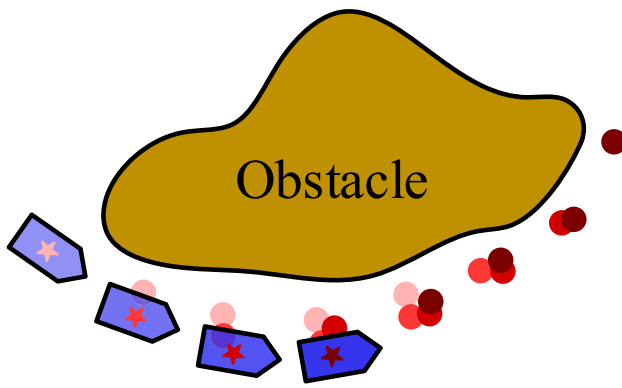


Fig. 9 The alternate mechanism of the SPC method

planning and control (SPC) method is developed to address the above issues.

In the developed SPC method, as illustrated in Fig. 9, the proposed IAPF and the MPC cooperate alternately. The IAPF is implemented for trajectory planning in a short future (the dots of the same color in Fig. 9) and The MPC is applied for trajectory tracking. The computed control command is applied to the USV, and the new state is fed back into the IAPF method to continue the iteration process.

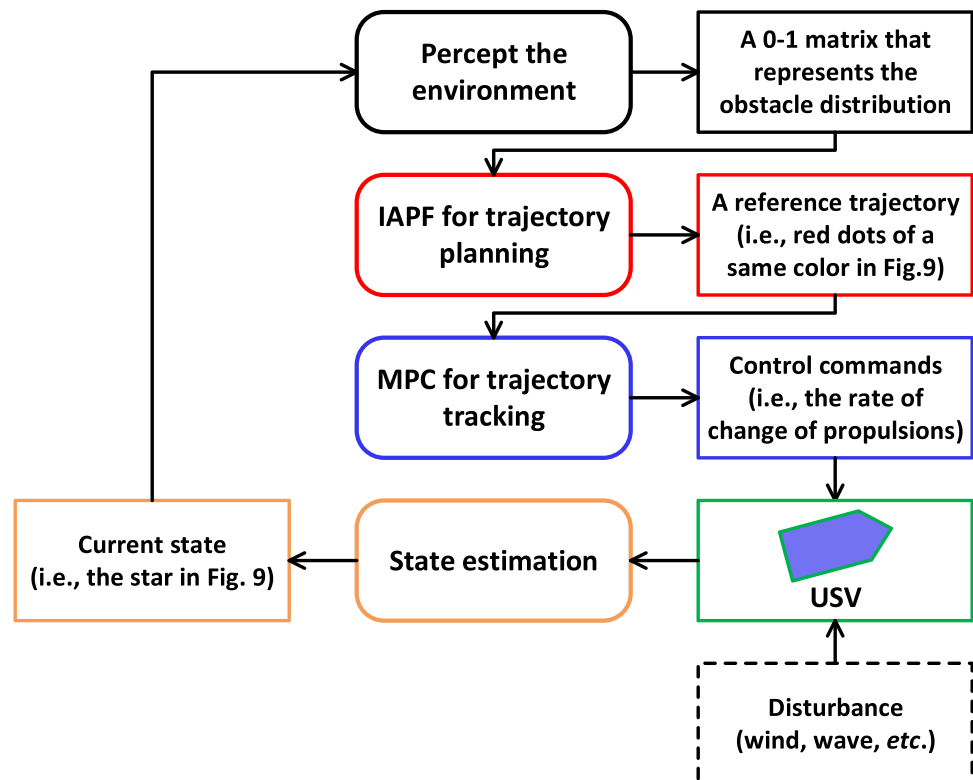
4.2 Design of the MPC

In the MPC, the sampling period and the prediction window length are fixed as δT and T , respectively. It is required that $M = T/\delta T \in \mathbb{Z}_+$ in the MPC and $m = T/\Delta T \in \mathbb{Z}_+$ in the IAPF. At the k th sampling instant t_k , the MPC solves an open-loop optimal control problem $P^{[k]}$ that defined over the time interval $[t_k, t_k + T]$. The problem $P^{[k]}$ is formulated as

$$P^{[k]} : \begin{cases} \min J = \int_{t_k}^{t_k+T} \|\hat{\mathbf{x}} - \hat{\mathbf{x}}_{ref}\|_Q^2 + \|u\|_R^2 \\ \text{s.t.} \\ \dot{\mathbf{x}} = f(\mathbf{x}, u, t), \\ \mathbf{x}(t_k) = \tilde{\mathbf{x}}_k \\ F_{\min} \leq F_i \leq F_{\max}, (i = 1, 2) \\ |u_i| \leq u_{\max}, (i = 1, 2) \\ 0 \leq v \leq v_{\max} \\ |\omega| \leq \omega_{\max} \end{cases} \quad (17)$$

where $\hat{\mathbf{x}} = (x, y, \theta)^T$ is the reduced state space of $\mathbf{x}$ and one has $\hat{\mathbf{x}} = \mathbf{A}\mathbf{x}$ with $\mathbf{A} = [\mathbf{I}_{3 \times 3} \mathbf{0}_{3 \times 4}]$; $\hat{\mathbf{x}}_{ref}$ is the reference trajectory generated by the IAPF over the time interval $[t_k, t_k + T]$, and it is a spline function that constructed based on the state series $\{\hat{\mathbf{x}}(t_k + j \cdot \Delta T)\}_{j=0}^m$; $\tilde{\mathbf{x}}_k$ is the measured state at the sampling instant t_k ; $\mathbf{Q} = \text{diag}(q_1, q_2, q_3)$ and $\mathbf{R} = \text{diag}(r_1, r_2)$ are the weighted matrices on tracking error and control inputs, respectively. $v_{\max}$ is the maximum allowable velocity; $\omega_{\max}$ is the maximum allowable amplitude of angular velocity.

Fig. 10 Flowchart diagram of the SPC method



4.3 Overall Design

According to the procedure given in Fig. 10, the IAPF and the designed MPC work alternatively in the SPC method. And the method terminates until the distance between the goal position and the USV current position is less than a prescribed tolerance ε . Finally, the pseudocode of the SPC method is given in Algorithm 1.

Algorithm 1: The simultaneous planning and control method	
Input:	
	The target position G ;
	USV properties, including m_c , I_c , r and r_{per} ;
	The approaching tolerance ε .
Output:	
	Optimal trajectory and corresponding control commands.
1.	Initialize parameters in the IAPF, including ΔT , v_{const} , $\tilde{\omega}$, K_{att} , K_{rep} , ρ_g , ρ_o and n .
2.	Initialize parameters in the MPC, including δT , T , $\mathbf{Q}$ and $\mathbf{R}$.
3.	Calculate variables $M = T/\delta T$ and $m = T/\Delta T$.
4.	Denote the current time t_0 ; set iteration index $k = 0$.
5.	while true do
6.	Record the current time $t_i = t_0 + k \cdot \delta T$.
7.	Obtain the current state of the USV $\tilde{\mathbf{x}}_k$, and extract the current position P_k .
8.	if $ \tilde{P}_k G \leq \varepsilon$ (i.e., USV approaches the target)
9.	break
10.	end if
11.	Implement the IAPF for m times to obtain a reference trajectory $\{\tilde{\mathbf{x}}(t_i + j \cdot \Delta T)\}_{j=0}^m$.
12.	Apply the MPC in Eq. (17) to track the reference trajectory.
13.	Update $k = k + 1$.
14.	end while

Remark 2 It is seen that the computational time consumed at each sampling instants consists of two parts, i.e., trajectory planning by the IAPF method and the following trajectory tracking by the MPC method. When implementing the IAPF, the computation of resultant repulsive force takes up much of the time, and the computational time is linear to the number of occupied cells. Hence, if one describes an obstacle by its contour rather than a concrete pattern, the computational time at trajectory planning will significantly decrease. However, one should carefully meet the balance between the thick of contour cells and the desired constant velocity of the USV; otherwise, the USV may penetrate into the obstacle. As for the MPC method, it is much more time-consuming since a nonlinear constrained optimization problem is required to be solved. On the one hand, it is recommended to adopt high quality computational optimal control numerical algorithm as the core solver, such as direct pseudospectral methods [32–34], symplectic pseudospectral methods [35–38]. On the other hand, high-quality initial guesses are expected to accelerate the nonlinear optimization process, and one can simply feed the convergent results in the current instant into the next instants as initial guesses.

5 Simulations

In this section, the developed SPC is validated by two examples. In the following simulations, properties of the USV are set as $m_c = 2\text{kg}$, $I_c = 0.25\text{kg} \cdot \text{m}^2$, $r = 0.05\text{m}$,

and $r_{per} = 20\text{m}$. Unless otherwise stated, the algorithm parameters used are set as follows. In the IAPF, we set $\Delta T = 0.2\text{s}$, $v_{const} = 1\text{m/s}$, $\tilde{\omega} = 20\text{deg/s}$, $K_{att} = 15$, $K_{rep} = 5$, $\rho_g = 500\text{m}$, $\rho_o = r_{per}$ and $n = 2$; As in the MPC, we set as $\delta T = 0.1\text{s}$, $T = 1\text{s}$, $\mathbf{Q} = \text{diag}(10^5, 10^5, 10^4)$, $\mathbf{R} = \text{diag}(1, 1)$. Then one has $M = 10$ and $m = 5$. As for constraints in the MPC, we set $v_{max} = v_{const}$, $\omega_{max} = \tilde{\omega}$, $F_{max} = -F_{min} = 2\text{N}$ and $u_{max} = 2\text{N/s}$. The optimal control problem is solved by the symplectic pseudospectral method (SPM) [39–41]. When implementing the SPM, the time horizon is divided into 2 sub-intervals with a 5-order Legendre–Gauss–Lobatto pseudospectral method applied in each sub-interval. All simulations are implemented using the MATLAB (R2021a) software on an Intel® Core™ i7-8550U machine with a 1.80 GHz processor and 8 GB of RAM.

5.1 Example 1

In the first example, we consider autonomous navigation in an obstacle-dense environment, as illustrated in Fig. 11. The initial state of the USV is set as $\mathbf{x}^{[0]} = (30\text{m}, 30\text{m}, 90\text{deg}, 0\text{m/s}, 0\text{deg/s}, 0\text{N}, 0\text{N})^T$, and three cases with different targets are considered. The map is decomposed into cells with the size of $1\text{m} \times 1\text{m}$. The computed trajectories for three different cases are summarized in Table 2. It is seen that the SPC method can successfully drive the USV to the prescribed target in each case, and the USV can successfully sail across narrow tunnels. Moreover, the constraints on angular velocity and propulsions are strictly satisfied.

Based on the map in Fig. 11, we also conduct 100 cases where the initial state and the prescribed target are randomly

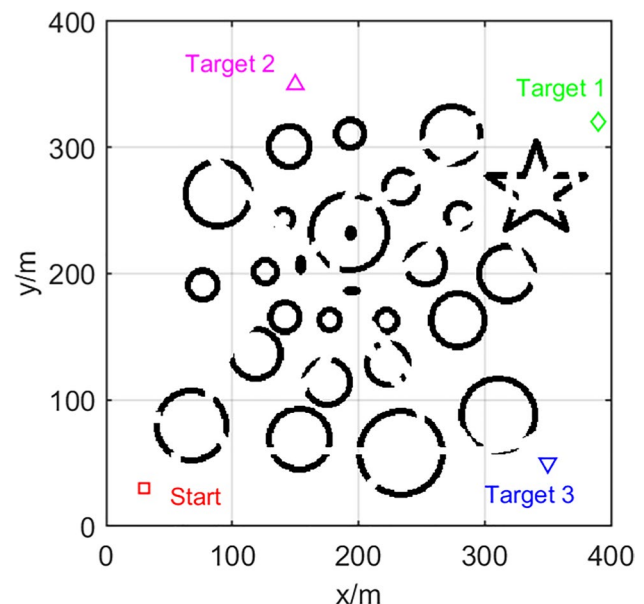
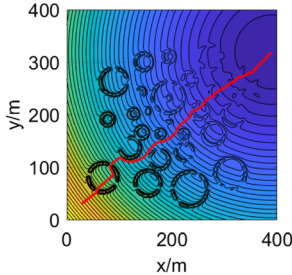
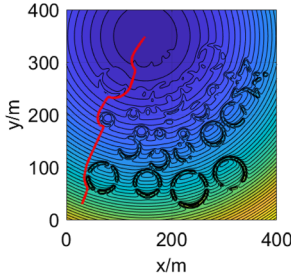
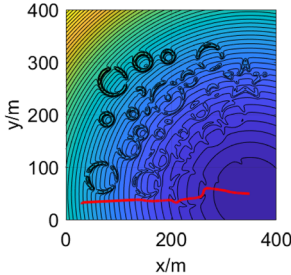
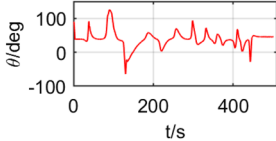
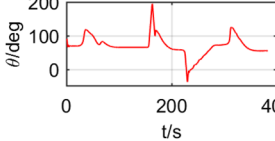
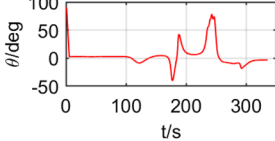
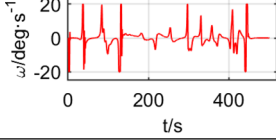
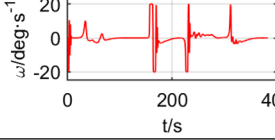
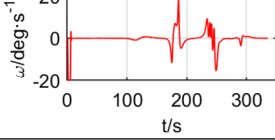
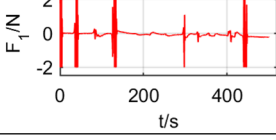
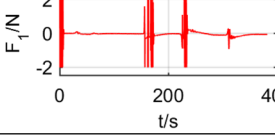
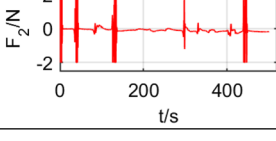
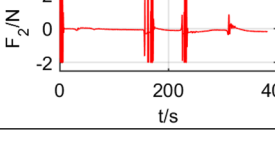
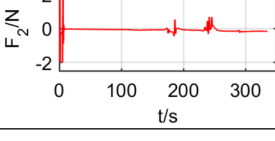


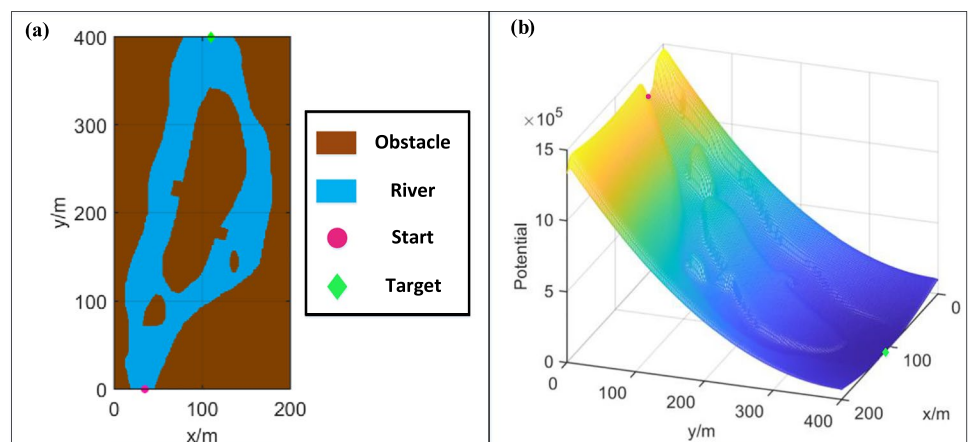
Fig. 11 Map in Example 1

Table 2 Trajectory results for three cases

	Case 1	Case 2	Case 3
Target	(390m, 320m)	(150m, 350m)	(350m, 50m)
Arrival time (s)	500.8	382.8	335.5
Trajectory and potential field			
Profiles of orientation (deg)			
Profiles of angular velocity (deg/s)			
Profiles of F_1 (N)			
Profiles of F_2 (N)			

generated. The DAPF and the IAPF are tested as the required APF module in the SPC framework. The success rates of the IAPF and the DAPF are 98% and 88%, respectively. The two cases where both IAPF and the DAPF fails are due to the irrationally set initial state. In these two cases,

the initial position of the USV is close to obstacles and the initial orientation of the USV is facing towards obstacles. It's not allowable for USV to make a sufficient turn maneuver before it crushes into obstacles. As for the ten cases where only the DAPF fails, the DAPF generates reference

Fig. 12 (a) The simulation environment and (b) corresponding potential field

trajectories with severe oscillations, making the following MPC procedure diverge. The above comparison highlights the advantage of the IAPF over the DAPF when implementing the SPC method.

5.2 Example 2

In this example, we consider a $200\text{m} \times 400\text{m}$ region as given in Fig. 12(a). At the initial time $t_0 = 0\text{s}$ the USV stops at $(35\text{m}, 0\text{m})$ with the orientation of 90° . Hence, the initial state of the USV is given as $\mathbf{x}^{[0]} = (35\text{m}, 0\text{m}, 90^\circ, 0\text{m/s}, 0^\circ/\text{s}, 0\text{N}, 0\text{N})^T$. And we set the target G as $(110\text{m}, 400\text{m})$. The environment is decomposed into cells with the size of $1\text{m} \times 1\text{m}$, and the corresponding potential field is plotted Fig. 11(b). Two cases are carried out to validate the effectiveness of the SPC method in practical scenarios. In the first case, no dynamic obstacles are distributed in the environment. As for the second case, we simulate the scenario where dynamic obstacles exist, and external disturbances are further considered.

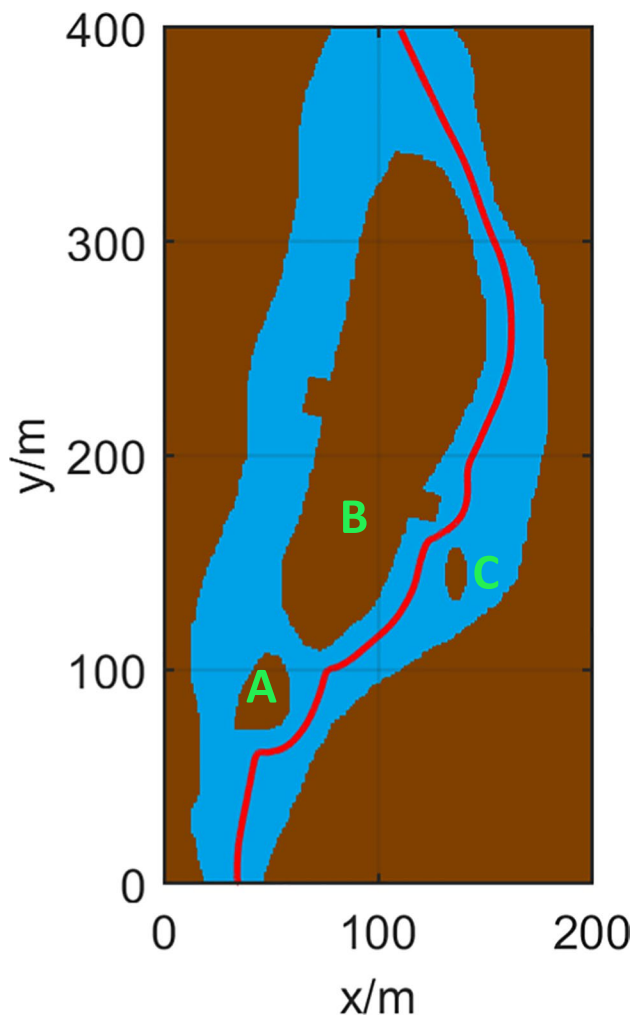


Fig. 13 Trajectory of the USV

5.3 Case 1: Scenario without Dynamic Obstacles

In this case, it takes the USV 460 s (i.e., 4,600 steps) to approach the target. The online computational time for each iteration is on average 4.29 ms, where 0.78 ms and 3.51 ms are consumed by the IAPF module and the MPC module, respectively. Compared to the given sampling interval $\delta T = 100\text{ms}$, the proposed SPC framework can be successfully applied to the autonomous navigation of USVs in real-time without apparent control delays. The actual trajectory of the USV is illustrated in Fig. 13. And profiles of state variables including position, orientation, velocity and angular velocity are given in Fig. 14. It is seen that evolutions of state variables are generally smooth while the imposed constraints on angular velocity are strictly satisfied.

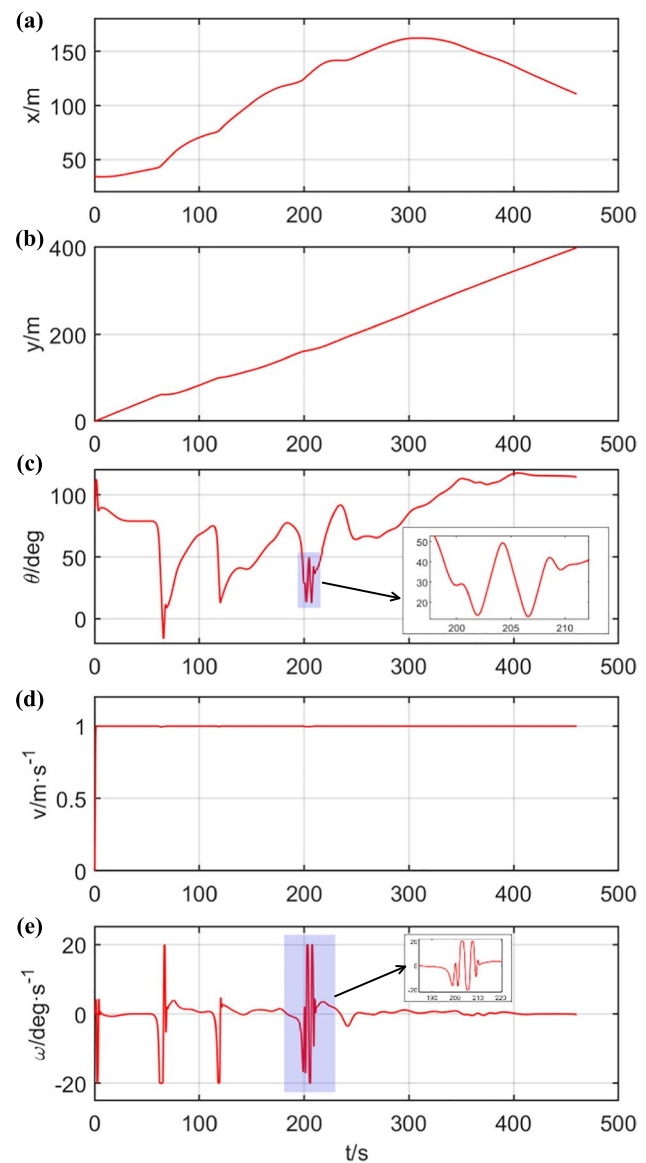


Fig. 14 Profiles of state variables

The evolutions of two propulsions are reported in Fig. 15. It is seen that the propulsions vary sharply during four time periods, which can be summarized and explained as follows:

- (1) At the very initial period, propulsions are required to accelerate the USV to the constant velocity. Additionally, since the USV is much closer to the east river band, the USV moves to the center of the river channel under the navigation of the IAPF module.
- (2) At about the 70 s the USV moves approximately northwards. After detecting reef A, the USV begin to move eastwards to detour the island B ahead.
- (3) At about 110 s, similar to the previous condition, the USV maneuver eastwards to turn around the corner of island B and then cruise along the contour of the island.
- (4) At approximately 195 s, the USV moves to the narrow river channel between island B and reef C. The potential field around this position is complicated, and the IAPF navigates the USV to adjust the course frequently, as seen in Fig. 14(c).

Though the propulsions are complicated within the aforementioned four time periods, they are successfully restrained within $[-2N, 2N]$ and the amplitudes of corresponding rates of change are kept within $u_{\max} = 2N/s$ during the whole control interval.

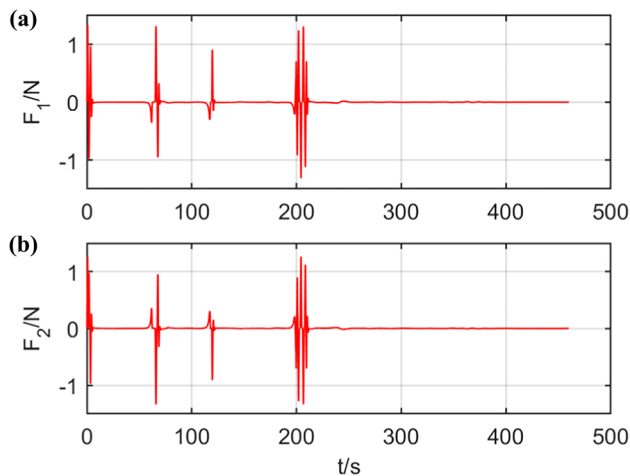


Fig. 15 Profiles of propulsions

Table 3 Motion information of two dynamic obstacles

No. Vessel	Initial position	Velocity	Moving direction
1	(40 m, 50 m)	1.0 m/s	Southwards
2	(30 m, 90 m)	1.5 m/s	Southwards

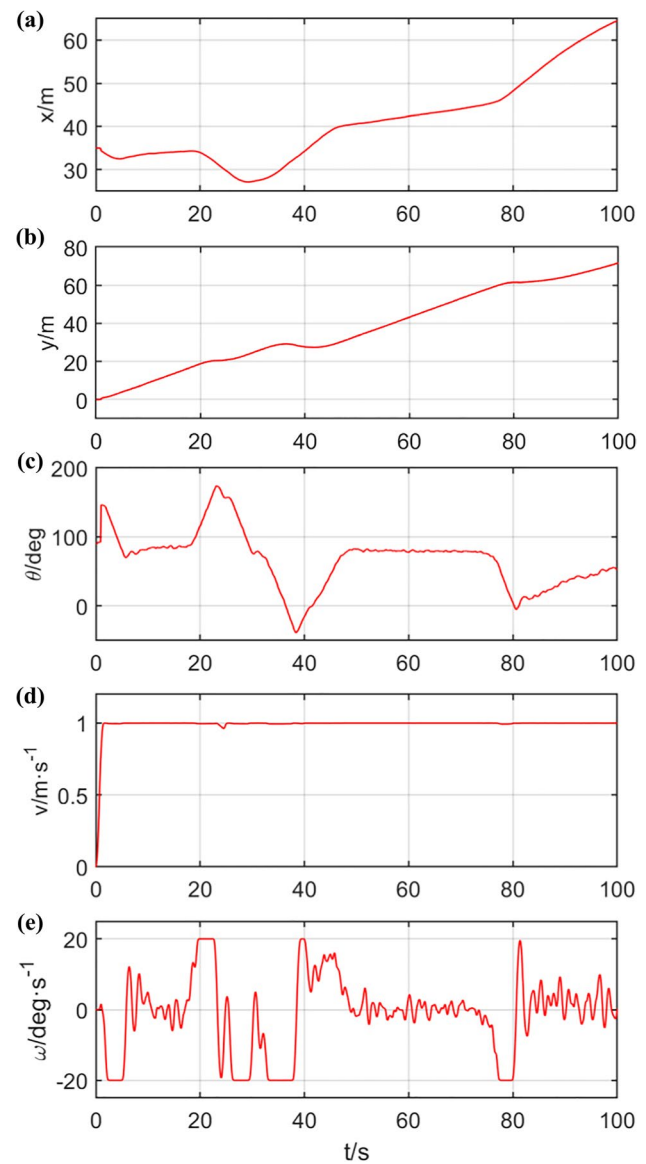


Fig. 16 Profiles of state variables

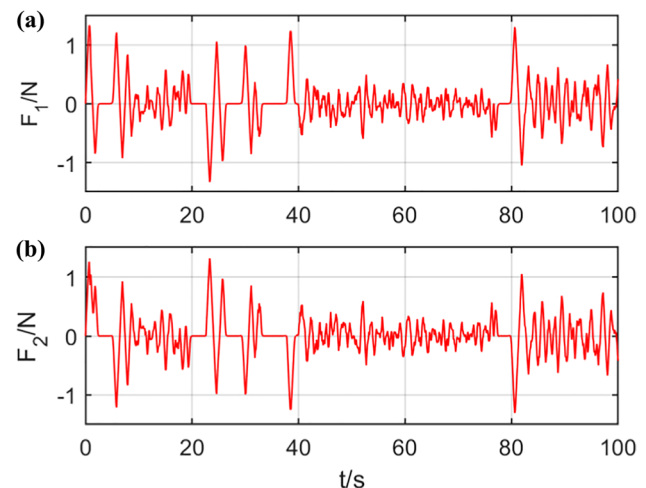


Fig. 17 Profiles of propulsions

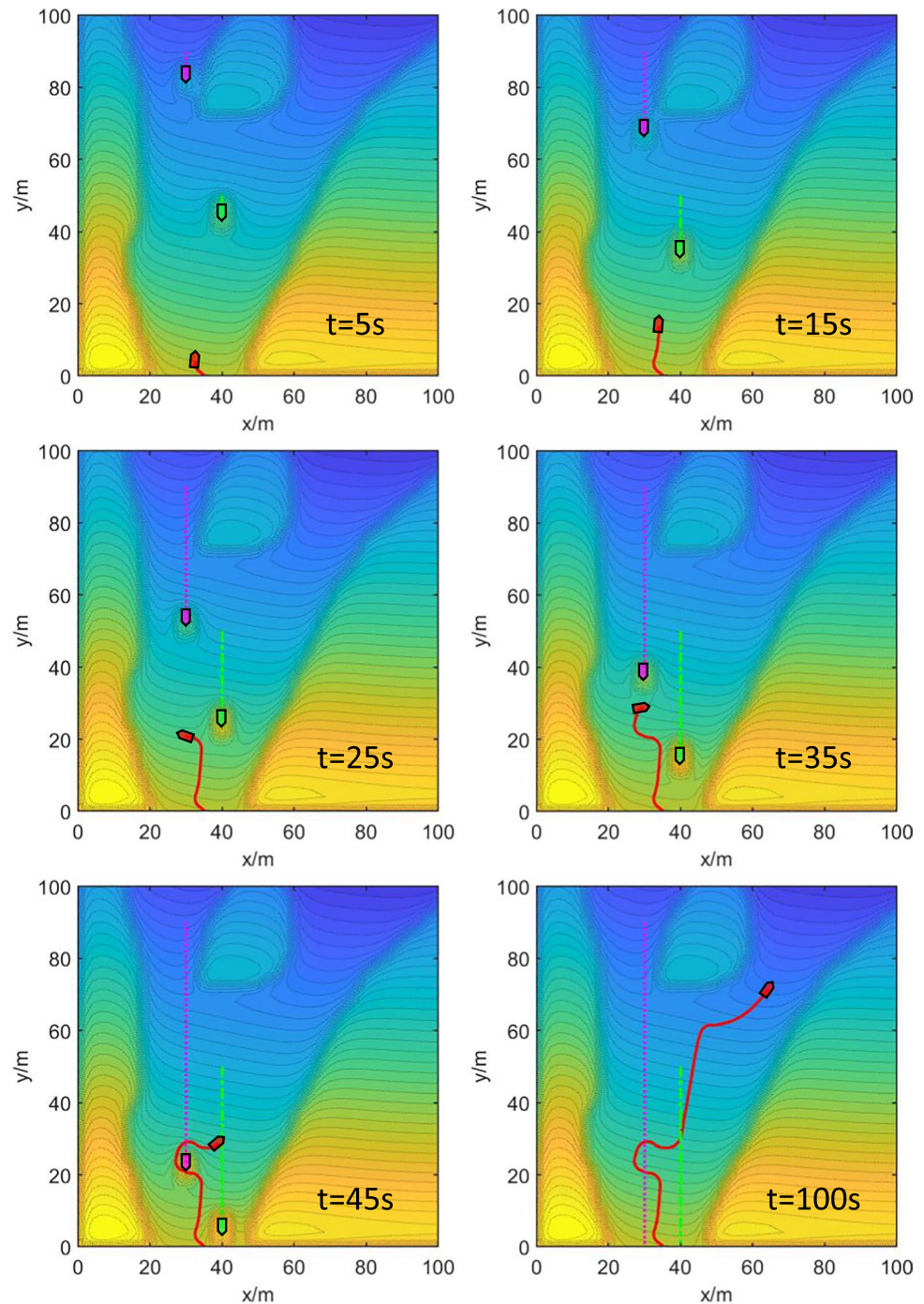
5.4 Case 2: Scenario with Dynamic Obstacles

Two vessels are introduced into the environment as dynamic obstacles and their motion information is reported in Table 3. Additionally, to consider the external disturbances, the input applied to the USV is multiplied by a coefficient $(1 + \lambda)$, where λ reflects the effect of disturbance that subjects to $\lambda \sim N(0, 0.01^2)$. To give a better illustration of the navigation process, only results within the first 100 s are reported. The profiles of state variables and control variables

are plotted in Fig. 16 and Fig. 17, respectively. Similar to those in Case 1, the evolutions of state variables are smooth and constraints on angular velocity are satisfied. However, due to the existence of external disturbances, the propulsions vary much more complicated.

In Fig. 18, the position and posture of the USV at different time instants (i.e., 5 s, 15 s, 25 s, 35 s, 45 s and 100 s) are illustrated. The contour of potential field at each time instant is set as the background. The red, green and pink vessels represent the position of the focused USV, obstacle-vessel 1 and obstacle-vessel 2, respectively. It is seen that

Fig. 18 Trajectory histories at different time instants



two obstacle-vessels generate extra potential peaks around their locations, thus influencing the motion of the USV. In approximately the first 15 s, the trajectory of the USV in this case is almost the same as that in Case 1. Then, the USV detects vessel 1 and decides to detour the obstacle by turning westwards. At 25 s, the USV has already avoided vessel 1 and detects vessel 2. At the influence of vessel 2 and the west river band, the USV decides to elude the obstacle by turning eastwards and basically achieve the avoidance of vessel 2 at 40 s. After that, the cruise path is almost the same as that in Case 1.

5.5 Discussions

From the above examples, it is seen that the developed SPC method can successfully drive the USV to the target by avoiding obstacles in the environment. At the same time, the dynamic motion equations and constraints on velocity, angular velocity, propulsions and rate of change of propulsions can be strictly satisfied. Compared with the pure APF method, optimization techniques are used to meet the above necessary constraints. Considering the practical working environment, especially where dense obstacles distributed, it is generally hard to directly apply the optimal control method. In the developed SPC method, the IAPF module generates a local reference trajectory in a short future, then a simple trajectory tracking problem without obstacle avoidance is left for the MPC to solve.

In general case, one can also use the IAPF method to generate a global reference trajectory and send it to MPC for trajectory tracking. However, generating such a global reference trajectory could be extremely time-consuming when the map is broad. And in extreme cases where large disturbance works, it becomes infeasible for the USV to accurately track the reference planned trajectory due to the mechanical limit. In our SPC method, the IAPF and the MPC work in an alternative manner, greatly eliminating the influence of external disturbances.

The designed SPC method drive the USV to the constant velocity since a constant desired velocity is used in the IAPF. Additionally, the final approach orientation cannot be prescribed. Hence, the SPC method cannot be directly applied to the autonomous docking task for USVs. There can be two ways to address these two issues. In the first manner, one can further develop the IAPF by introducing mechanisms on velocity adjusting and satisfaction on terminal orientation. As for the second manner, when USV approaches around the target, trajectory planning based on optimal control techniques can be applied to replace the IAPF method. The desired terminal orientation and terminal zero-velocity can be satisfied by imposing terminal boundary conditions in the formulated optimal control problem.

6 Conclusions and Future Research Direction

An SPC method is developed for autonomous navigation of underactuated USVs in unknown environments. By introducing constraints on angular velocity into the DAPF method, an IAPF method is reconstructed and used for trajectory planning. The MPC method is applied for trajectory tracking. The IAPF method and the MPC method cooperate alternately to navigate the USV to approach the target position. Simulations demonstrate the effectiveness of the SPC method in the presence of both static and dynamic obstacles. The SPC method is also applicable in actual environments with external disturbances due to the benefit of the robustness of MPC. In future research, as discussed in Section V.C, we will continue improving the SPC method to adopt it for autonomous docking. And cooperative autonomous navigation for multiple USVs will also be investigated.

Authors' Contributions Wang X and Liu J contributed to project conceptualisation. Wang X, Peng H and Zhao X built the model. Wang X, Qie X and Lu C designed the scenario settings. Wang X and Liu J wrote the manuscript. Liu J, Qie X and Lu C edited the manuscript. Liu J supervised the project. All authors approved the manuscript.

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Code Availability Not applicable.

Declarations

Ethics Approval This article does not contain any studies with human participants or animals performed by any of the authors.

Consent to Participate All authors agreed to participate the research.

Consent for Publication All authors read and approved the final manuscript.

Conflicts of Interest The authors declare no conflict of interests.

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